



Series €FGHE/C



Set-5

प्रश्न-पत्र कोड
Q.P. Code

56(B)

रोल नं.

Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

रसायन विज्ञान (सैद्धांतिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

CHEMISTRY (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 35 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 23 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 35 questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions carefully and strictly follow them :

- (i) This question paper contains **35** questions. **All** questions are **compulsory**.*
- (ii) This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*
- (iii) In **Section A** – Questions no. **1** to **18** are multiple choice (MCQ) type questions, carrying **1** mark each.*
- (iv) In **Section B** – Questions no. **19** to **25** very short answer (VSA) type questions, carrying **2** marks each.*
- (v) In **Section C** – Questions no. **26** to **30** are short answer (SA) type questions, carrying **3** marks each.*
- (vi) In **Section D** – Questions no. **31** and **32** are case-based questions carrying **4** marks each.*
- (vii) In **Section E** – Questions no. **33** to **35** are long answer (LA) type questions carrying **5** marks each.*
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 2 questions in Section E.*
- (ix) Use of calculators is **not** allowed.*

SECTION A

The following questions are multiple choice questions with one correct answer. Each question carries 1 mark. There is no internal choice in this section.

18×1=18

- 1.** A solution that obeys Raoult's law is called :
- (a) a non-ideal solution
 - (b) a molar solution
 - (c) an ideal solution
 - (d) a saturated solution



2. The products of the electrolysis of aqueous NaCl are :
- (a) H_2 and O_2
 - (b) H_2 , Cl_2 and NaOH
 - (c) H_2 , O_2 and Na
 - (d) O_2 and NaOH
3. Which of the following cells is **not** rechargeable ?
- (a) Dry cell
 - (b) Lead storage battery
 - (c) Ni-Cd battery
 - (d) Fuel cell
4. The unit of zero order reaction is :
- (a) s^{-1}
 - (b) $\text{L}^2 \text{mol}^{-1} \text{s}^{-1}$
 - (c) $\text{mol L}^{-1} \text{s}^{-1}$
 - (d) L s^{-1}
5. Which of the following transition metal ions is a strong oxidising agent in aqueous solution ?
- (a) Zn^{2+}
 - (b) Mn^{2+}
 - (c) Mn^{3+}
 - (d) Cr^{2+}
6. Which of the following ligands is an ambidentate ligand ?
- (a) OH^-
 - (b) NH_3
 - (c) CO
 - (d) NO_2^-



7. How many hydrate isomers are possible of the complex with the formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$?
- (a) Six (b) Four
(c) Three (d) Two
8. Which of the following, on reaction with CH_3MgI and subsequent hydrolysis, will form a primary alcohol ?
- (a) HCHO (b) CH_3CHO
(c) CH_3COCH_3 (d) Benzaldehyde
9. Which of the following compound is called Picric acid ?
- (a) p-Nitrophenol
(b) 2,4-Dinitrophenol
(c) 2,4,6-Trinitrophenol
(d) 2,6-Dinitrophenol
10. 1,2-Dichloroethane is heated with KOH (1 mole) in ethanol. The major product formed is :
- (a) Acetylene
(b) Vinyl chloride
(c) 2-Chloroethanol
(d) Ethylene glycol
11. The osmotic pressure of a solution increases if the :
- (a) number of solute molecules is increased.
(b) temperature is decreased.
(c) volume is increased.
(d) number of solute molecules is decreased.



12. Clemmensen Reduction of aldehydes and ketonic carbonyl groups into methylene group is effected by the reagent :
- (a) H_2 , Pd (b) LiAlH_4
(c) Zn-Hg , conc. HCl (d) $\text{NH}_2 - \text{NH}_2$, KOH
13. Vitamin B_{12} is also called :
- (a) Riboflavin (b) Thiamine
(c) Pyridoxine (d) Cyanocobalamine
14. Which one among the following bases is present in RNA but usually **not** in DNA ?
- (a) Uracil (b) Thymine
(c) Adenine (d) Guanine

For Questions number 15 to 18, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.
15. ***Assertion (A) :*** Ammonolysis of alkyl halides is not a suitable method for the preparation of pure primary amines.
- Reason (R) :*** Ammonolysis of alkyl halides yields mainly tertiary amines.



- 16.** *Assertion (A) :* Sucrose is a non-reducing sugar.
Reason (R) : Sucrose has a glycosidic linkage.
- 17.** *Assertion (A) :* Zn, Cd and Hg are not regarded as transition elements.
Reason (R) : Zn, Cd and Hg have completely filled d orbitals (d^{10}) in their ground state as well as in their common oxidation state.
- 18.** *Assertion (A) :* Hydrolysis of an ester in acidic medium is a second order reaction.
Reason (R) : For a complex reaction, order is given by the slowest step.

SECTION B

This section contains 7 questions with internal choice in two questions. The following questions are very short answer type and carry 2 marks each :

$7 \times 2 = 14$

- 19.** What type of deviation from Raoult's law is shown by ethanol and acetone mixture ? Give reason. 2
- 20.** (a) Write anode, cathode and overall reaction of fuel cell. 2

OR

- (b) (i) State Faraday's first law of electrolysis.
- (ii) Why does the conductivity of a solution decrease with dilution ? $2 \times 1 = 2$
- 21.** A reaction is second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is : $2 \times 1 = 2$
- (i) doubled ?
- (ii) reduced to half ?



22. (a) Write the IUPAC names of the following coordination compounds : 2×1=2



OR

(b) What is spectrochemical series ? What is the difference between a strong field ligand and a weak field ligand ? 2

23. What happens when : 2×1=2

(a) Isopropyl chloride is treated with Na in dry ether ?

(b) Ethyl chloride is treated with AgCN ?

24. How will you convert ethanal into :

(a) But-2-enal

(b) Ethanol 2×1=2

25. Write the reaction involved in : 2×1=2

(a) Hoffmann-bromamide reaction, and

(b) Carbylamine reaction

SECTION C

This section contains 5 questions with internal choice in two questions. The following questions are short answer type and carry 3 marks each. 5×3=15

26. Calculate the boiling point of solution when 4 g of MgSO_4 ($M = 120 \text{ g mol}^{-1}$) was dissolved in 100 g of water, assuming MgSO_4 undergoes complete dissociation.

(K_b for water = $0.52 \text{ K kg mol}^{-1}$)

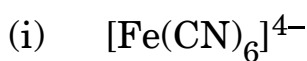


27. The rate constant of a first order reaction increases from $2 \times 10^{-2} \text{ s}^{-1}$ to $4 \times 10^{-2} \text{ s}^{-1}$ when the temperature changes from 300 K to 310 K. Calculate the energy of activation (E_a). 3

[$\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 4 = 0.6021$]

[$2.303 R = 19.15 \text{ JK}^{-1} \text{ mol}^{-1}$]

28. (a) Write the hybridisation and magnetic character of the following complexes : $3 \times 1 = 3$



[Atomic number : Fe = 26, Co = 27, Ni = 28]

OR

- (b) (i) Write the electronic configuration of Fe^{3+} on the basis of crystal field theory when it forms an octahedral complex in the presence of (i) strong field ligand, and (ii) weak field ligand.

[Atomic number : Fe = 26]

- (ii) $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic. Why ? 3

29. Answer any **three** of the following : $3 \times 1 = 3$

- (a) Out of -Cl and -CH₂-Cl, which one is more reactive towards $\text{S}_\text{N}2$ reaction and why ?
- (b) Out of chlorobenzene and p-nitrochlorobenzene which is more reactive towards nucleophilic substitution reaction and why ?
- (c) Out of 2-Bromopentane and 1-Bromopentane which one is optically active and why ?
- (d) Define ambident nucleophile with an example.



30. What happens when aniline is treated with 3×1=3
- (a) Br₂ water ?
 - (b) Conc. HNO₃ ?
 - (c) CH₃COCl/Pyridine ?

SECTION D

The following questions are case-based questions. Each question has an internal choice and carries 4 (1 + 1 + 2) marks each. Read the case carefully and answer the questions that follow. 2×4=8

31. Alcohols have higher boiling points than other classes of compounds, namely hydrocarbons, ethers and haloalkanes of comparable molecular masses.

Alcohols and phenols are acidic in nature. Electron withdrawing groups in phenol increase its acidic strength and electron releasing groups decrease it. Alcohols may be prepared by acid catalysed hydration of alkenes, hydroboration-oxidation reaction, from carbonyl compounds by catalytic reduction and the action of Grignard reagents.

Answer the following :

- (i) Why is *o*-nitrophenol more acidic than *o*-methoxyphenol ? 1
- (ii) Why does propanol have higher boiling point than propane ? 1
- (iii) Name the reagents used in the following reactions : 2×1=2
 - (1) Oxidation of primary alcohol to aldehyde
 - (2) Dehydration of propan-2-ol to propene

OR

- (iii) How do you convert 2×1=2
 - (1) Prop-1-ene to propan-1-ol ?
 - (2) Acetone to propan-2-ol ?



- 32.** Nucleic acids are the polymers of nucleotides which in turn consist of a base, a pentose sugar and phosphate group. Nucleic acids are responsible for the transfer of characters from parents to offsprings. There are two types of Nucleic acids — DNA and RNA. DNA is the chemical basis of heredity and have the coded message for proteins to be synthesised in the cell.

Answer the following :

- (i) What are nucleic acids ? 1
- (ii) What is the difference between Nucleoside and Nucleotide ? 1
- (iii) Write two differences between DNA and RNA. 2×1=2

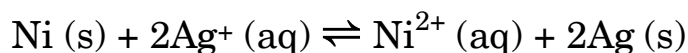
OR

- (iii) (1) The two strands in DNA are not identical but are complementary. Why ?
- (2) Write one important function of nucleic acid. 1+1=2

SECTION E

The following questions are long answer type and carry 5 marks each. Two questions have an internal choice. 3×5=15

- 33.** (a) Calculate the maximum work and $\log K_c$ for the given reaction at 298 K :



Given : $E^\circ_{\text{Ni}^{2+} / \text{Ni}} = -0.25 \text{ V}$, $E^\circ_{\text{Ag}^+ / \text{Ag}} = +0.80 \text{ V}$

$$1 \text{ F} = 96500 \text{ C mol}^{-1}$$



- (b) Explain the variation of molar conductivity with concentration for strong electrolyte and weak electrolyte with the help of diagram. 3+2=5

- 34.** (a) (i) Account for the following :
- (1) $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ is colourless.
 - (2) MnO is basic while Mn_2O_7 is acidic.
 - (3) Actinoids show a wide range of oxidation states.
- (ii) Write the preparation of KMnO_4 from MnO_2 . 3+2=5

OR

- (b) (i) Define lanthanoid contraction. Write its two consequences. Why is actinoid contraction greater than lanthanoid contraction ?
- (ii) Write the preparation of $\text{Na}_2\text{Cr}_2\text{O}_7$ from chromite ore. 3+2=5

- 35.** (a) (i) Write the products formed when Methanal reacts with the following :
- (1) Concentrated NaOH
 - (2) Semicarbazide/weak acid
 - (3) $\text{CH}_3\text{MgBr}/\text{H}_2\text{O}$
- (ii) Write the reaction involved in
- (1) Hell-Volhard-Zelinsky reaction, and
 - (2) Decarboxylation reaction 3+2=5

OR



- (b) (i) How will you convert the following ?
- (1) Toluene to Benzaldehyde
 - (2) Acetyl chloride to Ethanol
- (ii) Write a simple chemical test to distinguish between the following pairs of compounds :
- (1) Pentan-2-one and Pentan-3-one
 - (2) Benzoic acid and Benzaldehyde
- (c) Chloroacetic acid is more acidic than acetic acid. Why ?
- 2+2+1*